SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 0614 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

I Variganji Sumanth Babu / 192524447, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V. Sumanth Babu

Annexure A - Short Answer Test

QUESTION 1

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process. (5 Marks)



Functional blocks and their roles

1. Data collection: Gather representative examples from sensors, databases, logs, surveys, or images. The data must reflect the problem and the conditions in which the model will be used.

2. Preprocessing: remove duplicates and invalid records, handle missing values and outliers, encode categorical values, scale numeric features where needed, and split data without leakage.

3. Feature extraction/selection: convert raw inputs into useful predictors. For images these may be edges or learned embeddings; for tabular data they may be rainfall totals, age bands, or ratios. Remove redundant or irrelevant features using domain knowledge and validation.

4. Model training: fit model parameters on training data. A supervised model learns from example input-label pairs, while an unsupervised model searches for structure without target labels.

5. Evaluation and deployment: select thresholds and compare predictions with a separate validation/test set. Use metrics appropriate to the task, then monitor drift after deployment and retrain when the input distribution changes.

Supervised versus unsupervised learning

Aspect	Supervised learning	Unsupervised learning
Training data	Labelled pairs (x, y)	Unlabelled inputs (x)
Example	Email spam detection learns from emails marked spam/not spam.	Customer segmentation groups shoppers by purchase patterns without pre-assigned groups.
Output	Predicted class or numeric target	Clusters, embeddings, or latent structure

Conclusion: A reliable ML system is a repeatable pipeline, not just an algorithm. Data quality, leakage-free evaluation, and ongoing monitoring determine whether its predictions remain useful in practice.

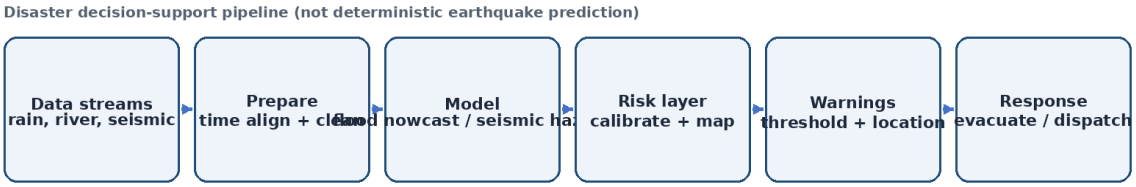
Worked example: predicting diabetes risk

A clinic could collect age, glucose, blood pressure, lifestyle information, and a verified outcome from historical visits. In supervised learning, each record contains a label such as disease present/not present, so the model learns a mapping from patient features to that label. The evaluation set must contain patients not used in training; splitting individual visits from the same patient across both sets can make results look artificially strong.

For an unsupervised analysis, the outcome label is withheld and patients are grouped by similarity of their measured profiles. These clusters may reveal patterns for further clinical investigation, but they are not diagnoses by themselves. In either approach, post-deployment monitoring checks for changes in measurement practices, patient populations, and error rates.

QUESTION 2

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response. (5 Marks)



System design and model choice

Use separate but connected models for different hazards because the physical signals and forecast horizons differ. For floods, start with Random Forest or gradient-boosted trees for structured features; use an LSTM/GRU or temporal convolution model when long rainfall, river-level, and soil-moisture sequences are available. For earthquakes, the defensible goal is seismic hazard/aftershock probability and impact assessment. Current science cannot reliably predict the exact time, location, and magnitude of an earthquake as a deterministic event; the system should communicate probabilities and uncertainty, not promise exact prediction.

Data preparation and feature selection

Combine historical incident records with rainfall forecasts, river gauges, soil moisture, topography, drainage, land use, seismic sensors, and fault-zone information. Synchronize timestamps and geographic units, remove duplicates, flag sensor faults, impute missing values carefully, handle outliers, encode categories, and scale inputs for neural networks. Candidate flood features include 1/6/24-hour rainfall, river level and rate of rise, antecedent wetness, slope, elevation, and distance to a river. Seismic features may include measured ground motion, event density, and site vulnerability. Select features using domain knowledge, mutual information or model-based importance, and validation-only comparisons.

Training, evaluation, and response

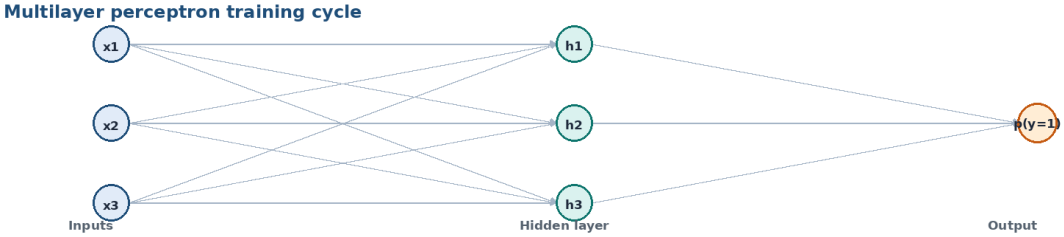
Split data chronologically and, where possible, hold out locations so nearby records do not leak into both train and test sets. Tune on training/validation data using time-series or grouped cross-validation; reserve the test set for final assessment. Report precision, recall, F1, PR-AUC for rare events, calibration, false-alarm rate, missed-event rate, and warning lead time. Compare with a simple baseline and test across seasons and extreme events. Convert calibrated risk to geographic alert levels, route warnings through official channels, identify exposed communities and evacuation routes, and display uncertainty to operators. Human authorities remain responsible for issuing official warnings.

Practical value: the system supports earlier, better-targeted decisions; its safety depends on sensor reliability, honest uncertainty, robust validation, and a clear fallback when data streams fail.

Operational safeguards are part of the design, not an afterthought. Authorities should set alert thresholds with emergency-management experts: a lower threshold may catch more dangerous events but also trigger more false alarms, while a higher threshold can miss events. Maps should show the affected area, likely lead time, confidence level, and recommended action. The system should be tested through historical replays and simulated outages, maintain backup communication channels, and record which alerts were acted upon. After each event, compare forecasts with observations and review false positives and missed events before updating the model.

QUESTION 3

3. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model. (5 Marks)



Steps in backpropagation

1. Forward pass: inputs are multiplied by weights and shifted by biases. Each layer computes z = W a + b and applies an activation function a = f(z). A sigmoid output gives p, the estimated probability of the positive class.

2. Error calculation: for binary classification use binary cross-entropy L = -[y log(p) + (1-y) log(1-p)]. If the true label is y=1 and the output is p=0.70, the prediction is not perfect; the loss measures the remaining error.

3. Backward pass: the output error is propagated toward earlier layers using the chain rule. With sigmoid + binary cross-entropy, the output delta is δout = p-y, so this example gives δout = -0.30. A hidden-unit delta is δj = f'(zj) Σk(wjk δk).

4. Weight update: compute gradients and perform gradient descent: Wnew = Wold - η ∂L/∂W; bnew = bold - η ∂L/∂b. Repeat over mini-batches and epochs until validation performance stops improving.

Learning rate and activation functions

The learning rate η controls step size. If it is too large, loss may oscillate or diverge; if too small, learning is slow. Learning-rate schedules or optimizers such as Adam can help, but validation must guide tuning. ReLU is often effective in hidden layers because it reduces saturation for positive inputs; sigmoid is suitable for a single binary probability output. Sigmoid/tanh hidden units can saturate and produce vanishing gradients. Proper initialization, input scaling, and normalization improve stability.

Summary: backpropagation calculates how each parameter contributed to the loss; gradient descent uses those derivatives to adjust weights in the direction that reduces the loss.

Illustrative update: with y=1, p=0.70, sigmoid plus binary cross-entropy gives an output-layer error term p-y = -0.30. If the bias gradient equals this term and the learning rate is η=0.1, gradient descent updates the bias as bnew = bold - 0.1(-0.30) = bold + 0.03. The update pushes the output upward for this positive example. In a real multilayer network, each weight gradient also depends on the activation from the preceding layer; gradients are accumulated across a mini-batch before the optimizer applies an update.

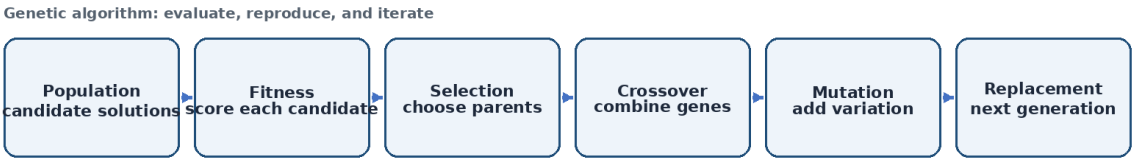
Training repeats this cycle over many batches. The training loss shows how well the current parameters fit seen examples, while validation loss estimates performance on unseen examples. If training loss continues to fall but validation loss rises, use early stopping or regularization rather than training indefinitely.

QUESTION 4

4. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution. (5 Marks)

Given population

Example	Speed	Accuracy	Fitness value	Suitable?
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes



Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution.

Applying the genetic algorithm

1. Fitness evaluation: use the supplied fitness value to rank candidates. Candidate 5 scores 80, candidate 3 scores 75, candidate 4 scores 65, candidate 2 scores 60, and candidate 1 scores 40.

2. Selection: a roulette-wheel or tournament strategy gives higher-scoring candidates a greater chance to become parents, while keeping some diversity so lower-ranked but useful traits are not immediately lost.

3. Crossover: represent each candidate as a two-gene chromosome [Speed, Accuracy]. For example, combining candidate 3 [High, Medium] with candidate 5 [Medium, High] can create offspring such as [High, High] or [Medium, Medium], depending on the crossover point. The offspring must then be re-evaluated.

4. Mutation and replacement: occasionally change one gene, such as Medium speed to High, to explore combinations missed by crossover. Keep the best individuals through elitism, score the new generation, and repeat until fitness stops improving or a generation limit is reached.

Best listed candidate: Example 5 (Speed = Medium, Accuracy = High, fitness = 80, Suitable = Yes). The dataset does not define chromosome encoding, exact crossover/mutation rates, or the fitness of offspring, so it supports identifying the best of the five given solutions, but not proving a global optimum after simulated evolution.

Generation 0: Rank the supplied candidates by fitness: Example 5 (80), Example 3 (75), Example 4 (65), Example 2 (60), and Example 1 (40). Select Examples 5 and 3 as parents because they have the two highest scores.

Generation 1 - crossover: Represent each candidate as [Speed, Accuracy]. Crossing [Medium, High] with [High, Medium] could produce [Medium, Medium] and [High, High], depending on the crossover point. These are illustrative offspring, not measured results.

Mutation and replacement: With a small mutation probability, change one attribute of an offspring to explore another combination. Evaluate every offspring with the actual fitness function, then form the next population while retaining the best candidate through elitism.

Result supported by the supplied data: Example 5 remains the best observed candidate with fitness 80 and Suitable = Yes. Because no fitness formula, chromosome encoding, or operator rates are given, numeric offspring scores and a proven global optimum cannot be claimed.

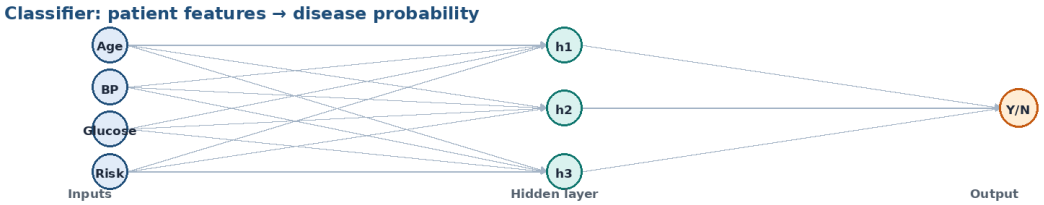
QUESTION 5

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. (5 Marks)

Supplied dataset

Ex.	Age	Blood pressure	Glucose	Lifestyle risk	Actual	Predicted
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.



Confusion matrix (positive class = Disease: Yes)

	Predicted Yes	Predicted No
Actual Yes	TP = 2	FN = 1
Actual No	FP = 1	TN = 1

Metrics and interpretation

TP = 2 (examples 1, 4); FN = 1 (example 2); TN = 1 (example 3); FP = 1 (example 5). Correct predictions = TP + TN = 3 of 5.

Accuracy = (TP+TN)/N = 3/5 = 60%. Precision = TP/(TP+FP) = 2/3 = 66.7%. Recall/Sensitivity = TP/(TP+FN) = 2/3 = 66.7%. Specificity = TN/(TN+FP) = 1/2 = 50%. F1 = 2TP/(2TP+FP+FN) = 4/6 = 66.7%.

The five-row sample indicates one missed disease case and one false alarm. These metrics are illustrative, not reliable estimates of real clinical performance: the sample is too small. Before training, encode categorical features appropriately, scale age and numeric measurements, impute missing values, inspect class balance, and select features using training data only. Evaluate on a larger independent, stratified dataset and prioritize sensitivity where missed disease is costly.

Training workflow: encode blood pressure and lifestyle categories (for example, one-hot encoding), normalize numeric measurements if the chosen network benefits from scaling, initialize weights, run the forward pass to obtain a disease probability, calculate binary cross-entropy, backpropagate gradients, and update weights over repeated epochs. Feature selection should remove leakage and redundant variables while retaining clinically justified predictors. The decision threshold should be checked against the cost of false negatives and false positives; only a separate validation set should guide this choice.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand